COMP5310 Assignment 1

Hotel booking cancellations

Dataset selection, data readiness and exploratory evidence

Group ID: 164

Contribution statement

Group member: SID and UniKey	Main responsibility and contribution	Agreed contribution (%)
520303830; texu0485	Worked on all components; discussed and consolidated each stage with the other member	50%
550278230; bsha0687	Worked on all components; discussed and consolidated each stage with the other member	50%

1 Dataset Audition and Selection

Table 1 compares the three supplied Canvas datasets. Only Hotel Booking receives detailed cleaning and EDA.

Table 1. Comparable lightweight dataset audition

Aspect	Airline Delay	Hotel Booking	Train Occupancy
Size and types	101,000 rows, 29 columns. Numeric, categorical, identifier, temporal and binary variables.	119,987 rows, 32 columns. Numeric, categorical, identifier, temporal and binary variables.	50,250 rows, 14 columns. Mainly categorical, identifier and temporal variables, plus stop order.
Outcome	Arrival delay at least 15 minutes: 19,440 of 97,700 observed outcomes (19.90%).	Cancellation/no-show: 44,448 of 119,987 booking records (37.04%); target complete.	Occupancy range: Low 91.41%, Medium 7.08%, High 1.52%.
Quality risks	3,300 missing target values; 1,000 potential duplicate rows; mixed formats. Delay-cause fields are mostly absent and would leak the outcome.	31,328 potential duplicate rows but no booking ID; missing values and category variants. reservation_status and reservation_status_date reveal the outcome.	250 potential duplicate rows; 73 missing status values; spelling/case variants. The outcome is highly imbalanced and coverage is short.
Decision	Airline or airport operations planner: allocate gates, turnaround buffers and disruption resources before departure.	Hotel revenue manager: decide which booking groups need attention when monitoring cancellations and planning room availability.	Rail operations planner: review staffing, passenger information and capacity for crowded stations and time bands.

Aspect	Airline Delay	Hotel Booking	Train Occupancy
Concern	January 2008 only; WN is 93.58% after trimming spaces and standardising case. Generalising could misallocate resources across carriers or routes.	Historical data from two hotels; country, channel and agent may act as proxy variables. Individual penalties based on cancellation risk could be unfair.	Only day values 9-16 and High occupancy is 1.52%. Decisions based on this short sample may disadvantage under-represented services.

Potential duplicate rows have identical values in all original fields, excluding the first occurrence. Hotel Booking has no booking ID, so these are not confirmed duplicate bookings.

Selection. Hotel Booking is the strongest Assignment 2 option: its target is complete, 44,448 of 119,987 records (37.04%) are cancellations/no-shows, and arrivals span July 2015-August 2017. Its main issues are missing values, category variants, potential duplicates and target leakage.

2 Stakeholder Problem and Success Criteria

The hotel revenue manager needs to decide which booking groups need greater attention when monitoring cancellations and planning room availability.

Research question: Which booking characteristics are associated with cancellation or no-show, and which booking groups have higher rates of these outcomes?

Target: is_canceled, where 1 means cancellation/no-show and 0 means check-out. In the raw dataset, 37.04% of bookings have a target value of 1. Throughout the analysis, cancellation rates include no-shows.

We exclude reservation_status and reservation_status_date as explanatory variables because they reveal the final outcome.

A useful answer should identify higher- and lower-rate groups, show booking and cancellation counts alongside rates, and check whether patterns hold within each hotel and across arrival months.

Cost of a misleading conclusion. Overestimating risk could waste staff time and inconvenience guests; underestimating it could leave the hotel unprepared for cancellations or no-shows and make room planning and resale more difficult.

3 Data Readiness Pipeline

Data dictionary

Notebook 02 contains the full dictionary for all 32 original fields plus arrival_date. Table 2 summarises the five fields used in A1 EDA; the cleaned CSV also retains 18 additional booking variables for possible later analysis.

Table 2. A1 EDA variables

Variable	Meaning / coding	Variable type
hotel	Hotel category	categorical
is_canceled	1 = cancellation/no-show; 0 = check-out	binary
lead_time	Days from booking to scheduled arrival	numeric count
market_segment	Booking market segment	categorical
arrival_date	Scheduled arrival date	temporal

The wider cleaned dataset also retains stay length, party size, booking/guest details, identifiers and requests for possible Assignment 2 analysis.

Audit of the original dataset

All 32 original columns are checked before cleaning for missing values, potential duplicates, data types, invalid values, inconsistent categories, unusual ranges and arrival dates.

- **Missing values:** children 182; meal 138; country 628; market segment 129; distribution channel 164; agent 16,541; company 113,162; customer type 154. The target has no missing values.
- Potential duplicates: 31,328 rows repeat all original field values. Dropping them would change the cancellation rate from 37.04% to 27.95%, but there is no booking ID to confirm duplicate bookings.
- **Types:** children is stored as float because it has missing values; agent and company are numeric-looking identifiers. Separate arrival fields need to be combined into a date.
- **Invalid values:** both binary fields contain only 0/1 and the checked count variables have no negative or fractional observations. There are 180 rows with known zeros for adults, children and babies, and one negative ADR value.
- **Categories:** variants include CityHotel, ResortHotel, bb, Direc, groups, offline ta/to, gds, contract and GROUP; country also uses CN alongside CHN.

- **Ranges and dates:** lead time has median 69 and maximum 1,636 days; ADR ranges from -6.38 to 5,400; 716 records have zero nights. All arrival dates are valid, spanning 1 July 2015 to 31 August 2017.

Cleaning process and impact

The pipeline removes bookings with no recorded guests, standardises selected labels, handles missing categories and identifiers, combines the arrival date, and saves the wider cleaned dataset.

hotel_bookings_clean.csv has 119,807 rows and 23 columns: five used in A1 EDA and 18 retained for possible later analysis. Only children remains missing (182 values). The cancellation/no-show rate is 37.08%, versus 37.04% before cleaning.

Table 3. Cleaning log

Issue and evidence	Action	Justification	Impact
180 rows with no recorded guests	Remove	All three guest counts are known and zero	119,987 -> 119,807 rows
Label variants, including CN country code	Standardise equivalent labels	Keep the same groups together and one country-code format	Hotel 4 -> 2 labels; CN -> CHN; other selected labels standardised
Undefined / missing categories and IDs	Meal Undefined -> SC; otherwise use Unknown	Keep known no-meal coding separate from unrecorded values	Missing selected categories/IDs handled without dropping rows
182 missing children counts	Keep missing as nullable integer	Unknown children counts are not zero	182 -> 182 missing
Separate arrival fields	Combine into arrival_date	Keep one usable scheduled arrival date	Separate components replaced by one date
31,328 potential duplicate rows	Retain	No booking ID confirms duplicate bookings	0 rows removed
Long lead times and zero-night records	Retain	No supported exclusion rule	Maximum lead time remains 1,636 days

Target leakage and availability at booking

reservation_status directly reveals the target: Canceled and No-Show records have a 100% cancellation rate, while Check-Out records have a 0% rate. reservation_status and reservation_status_date are excluded.

assigned_room_type, booking_changes and days_in_waiting_list are excluded because they are determined or accumulated after booking. ADR is also excluded because its booking-time meaning is unclear. The timing of retained fields should be confirmed before Assignment 2.

4 Exploratory Data Analysis

The three figures use the five A1 EDA columns from the wider cleaned dataset. Cancellation rate is the percentage of bookings with is_canceled = 1, including no-shows.

Lead time and cancellation

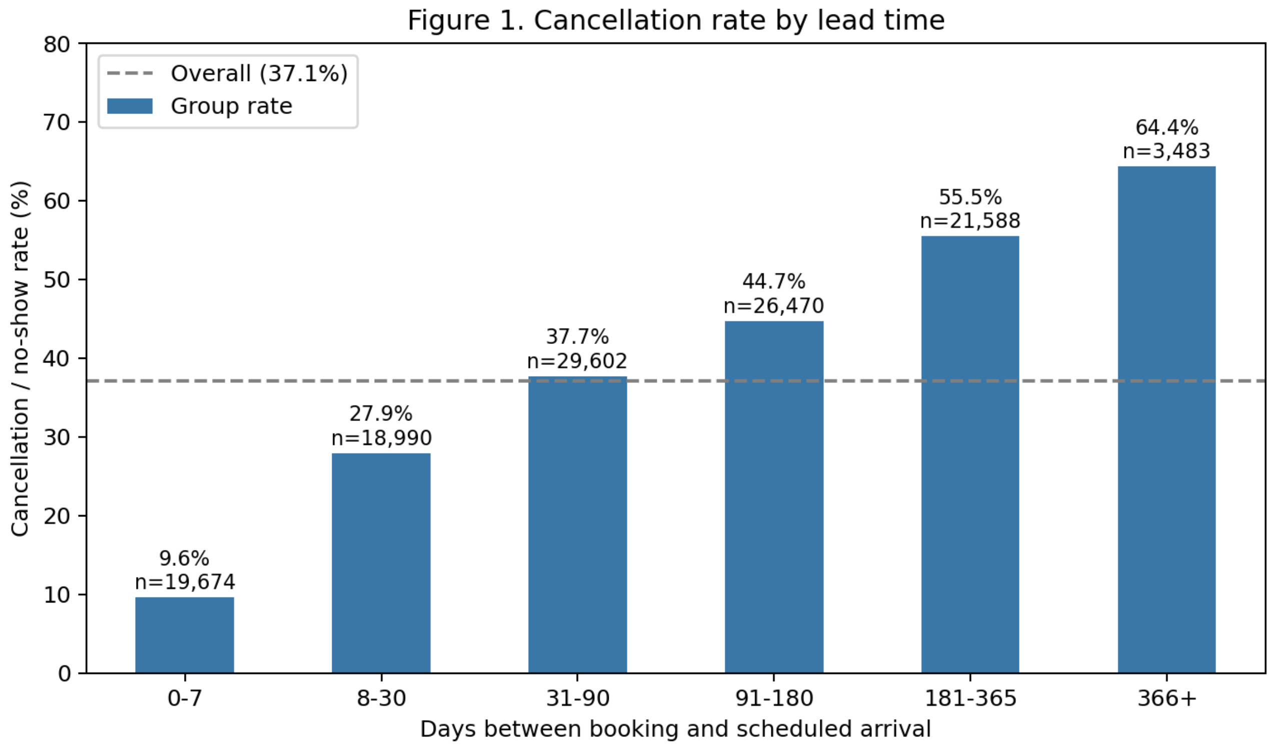

Figure 1. Lead times are grouped into readable bands; labels show cancellation rates and booking counts.

Rates rise from 9.6% at 0-7 days (19,674 bookings) to 64.4% beyond one year (3,483 bookings). Longer-lead bookings therefore deserve more attention in room planning. Assignment 2 should test whether this pattern remains after considering hotel, market segment and other retained booking details.

Market segment and cancellation

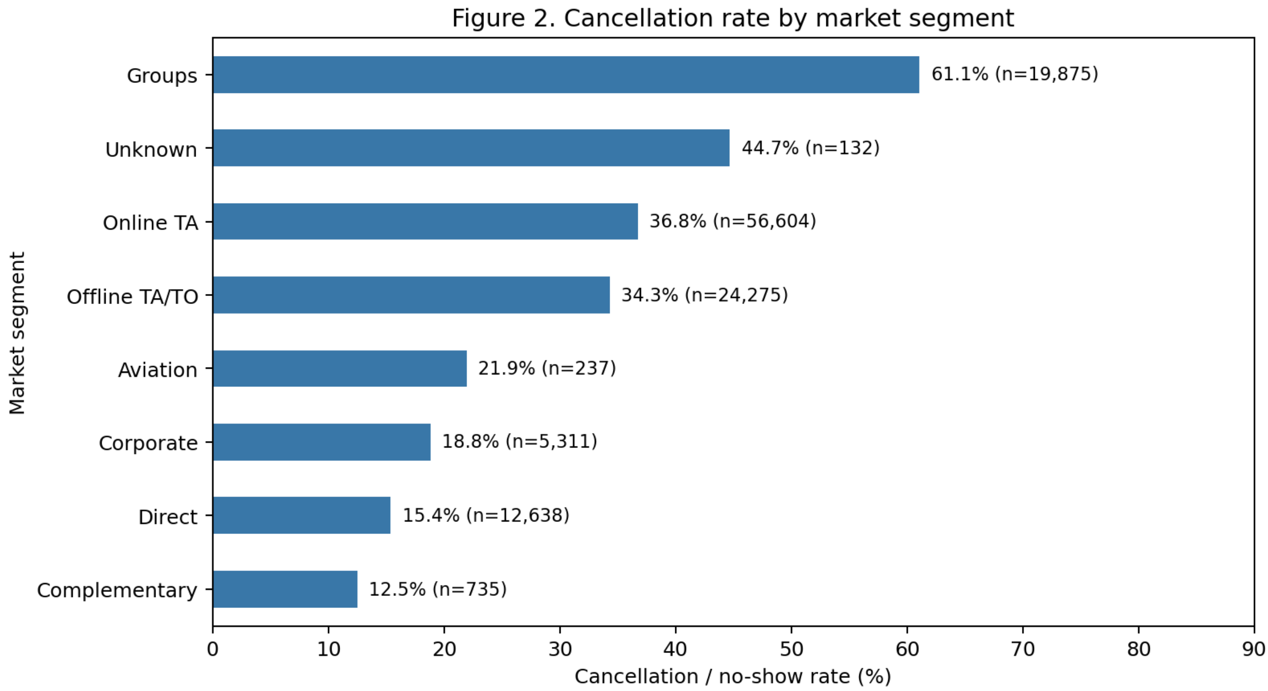

Figure 2. Cancellation rates and booking counts are shown for every market segment.

Groups has the highest rate (61.1%, 19,875 bookings), but Online TA produces more cancellations: 20,803 versus 12,142 for Groups. Both rate and booking volume matter for room planning.

Table 4. The two largest market segments within each hotel

Hotel	Market segment	Bookings	Cancellation rate
City Hotel	Offline TA/TO	16,775	42.8%
City Hotel	Online TA	38,798	37.4%
Resort Hotel	Offline TA/TO	7,500	15.2%
Resort Hotel	Online TA	17,806	35.2%

Offline TA/TO's overall rate is lower than Online TA's (34.3% versus 36.8%), but higher in City Hotel (42.8% versus 37.4%) and much lower in Resort Hotel (15.2% versus 35.2%). The manager should consider hotel and segment together.

Arrival month and hotel

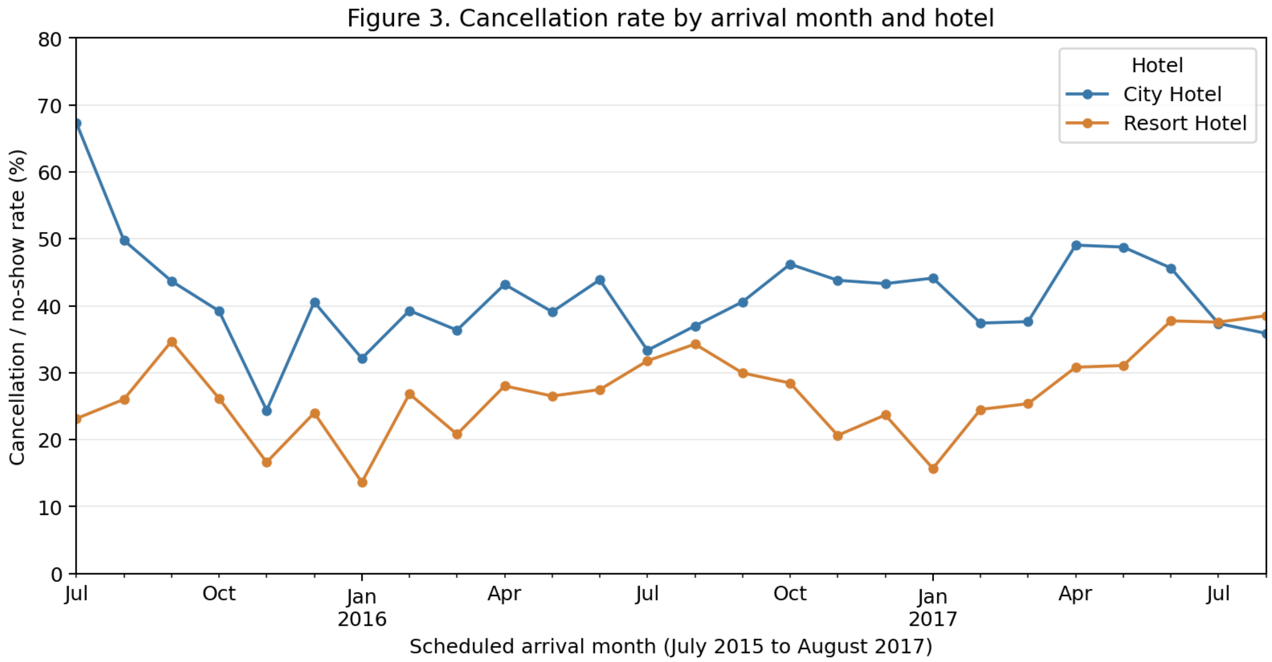

Figure 3. Rates are calculated by scheduled arrival month, keeping year and month together because the data run from July 2015 to August 2017.

City Hotel's overall rate is 41.8% versus Resort Hotel's 27.8%, but the gap changes over time. In August 2017, Resort Hotel reaches 38.5% (1,810 bookings), above City Hotel's 35.9% (3,141 bookings).

Room planning should reflect each hotel's recent patterns rather than apply the overall 37.08% rate everywhere. With 26 months and only one complete calendar year, the data do not establish a stable seasonal pattern.

Assignment 2 direction and limitations

- **Target and starting features:** predict is_canceled using lead time, hotel, market segment and arrival month as the starting features. Keep lead time numeric; the bands are only for Figure 1.
- **Additional candidates:** test the 18 extra retained variables only if they add useful information; do not automatically use every column.
- **Preparation and evaluation:** confirm booking-time availability, use a later-period holdout, encode categories and handle missing children using training data only. Compare precision, recall and a confusion matrix.
- **Models and questions:** start with logistic regression and consider a small decision tree. Test whether lead time remains useful after other booking details and whether hotel and segment should be modelled together.

- **Limitations:** potential duplicates and rare categories need care. Country or agency should not justify unfair treatment, and two historical hotels limit generalisation.

The fields excluded in the target-leakage section remain excluded from the proposed predictors. No predictive models are trained or compared in Assignment 1.

5 Reproducibility and Team Process

Reproducible handover

The README documents dependencies, file locations, run order and outputs. Place the three Canvas CSV files beside the notebooks and run each from top to bottom using relative paths:

1. 01_dataset_audition.ipynb: compares the three datasets and selects Hotel Booking.
2. 02_data_readiness.ipynb: audits and cleans Hotel Booking data, saves the 23-column cleaned CSV, and produces the three EDA figures and Assignment 2 plan.

Notebook 02 recreates the 23-column cleaned CSV and three EDA figures. The raw Canvas datasets are not included in the submission ZIP.

Week 6 role review

The group reviewed roles in Week 6 and retained the original 50/50 contribution split. Both members had contributed to the dataset selection, data preparation, EDA and written report, so no rebalancing was needed.

For cross-review, each member reviewed the other member's work before the final version was consolidated. This included checking the notebook outputs against the written report and confirming that the cleaning decisions, figures and interpretations were consistent.

References

Antonio, N., de Almeida, A. and Nunes, L. (2019) 'Hotel booking demand datasets', *Data in Brief*, 22, pp. 41-49. Available at: https://doi.org/10.1016/j.dib.2018.11.126.

The University of Sydney (2026a) *COMP5310 Assignment 1 datasets: NSW Train Occupancy, Air Travel Delay and Hotel Booking* [CSV datasets supplied through Canvas].

The University of Sydney (2026b) *Lab 3: Data Exploration with Python* [COMP5310 tutorial notebook, 03_data_exploration_with_Python.ipynb].

AI acknowledgement

OpenAI Codex assisted with code and report drafting and revision. Outputs were checked against the assignment requirements, course materials and observed data.